

2019 Airline Summary Dashboard

\$42K

Profit per Flight

\$78bn

Total Profit

\$111bn

Total Revenue

70%

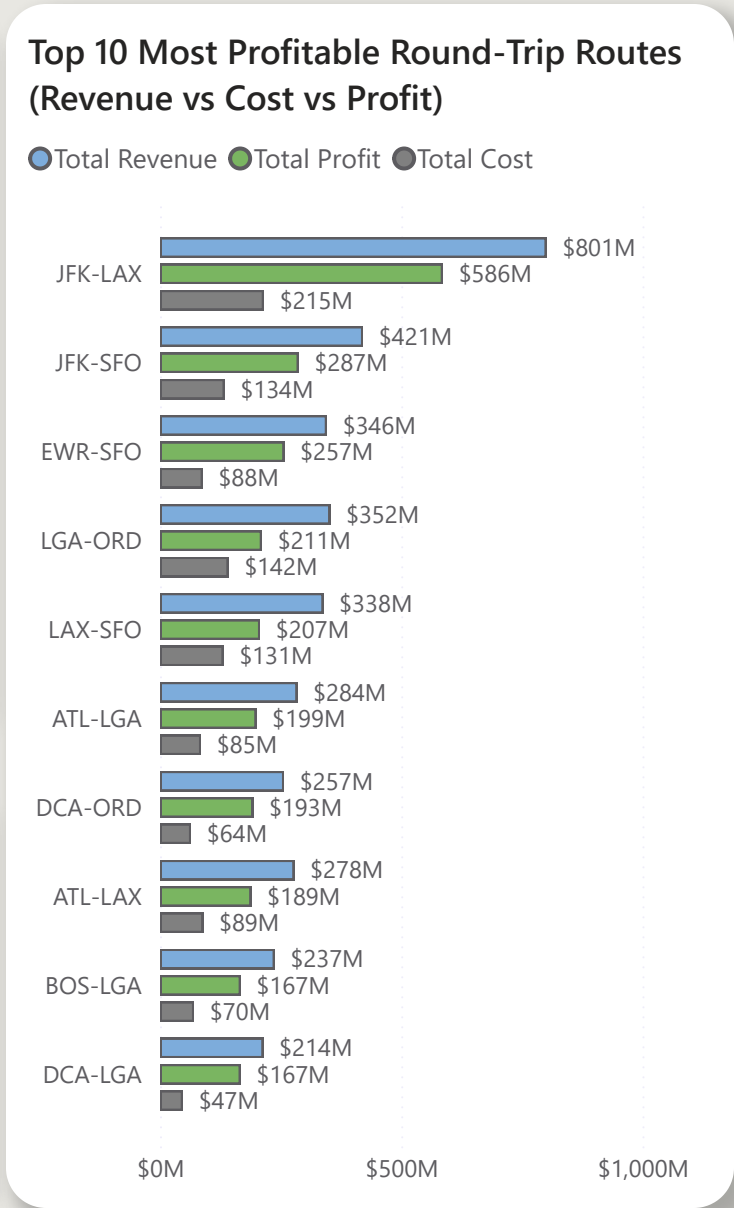
Profit Margin %

\$2K

Delay Cost per Flight

65%

Average Occupancy Rate



Aircraft Investment Breakeven by Recommended Round-Trip Route

Route	Profit per Flight	Breakeven Round Trip Flights	Total Cost	Total Profit	Total Revenue
SLC-TWF	\$264,533	378	\$6,284,033	\$153,958,122	\$160,242,155
CLT-FLO	\$252,272	396	\$4,681,395	\$127,145,276	\$131,826,671
GUM-HNL	\$180,425	554	\$7,990,868	\$31,935,202	\$39,926,070
PIH-SLC	\$165,918	603	\$5,260,380	\$89,595,833	\$94,856,213
HNL-PPG	\$118,204	846	\$1,714,076	\$6,383,028	\$8,097,105
Total	\$981,352	2777	\$25,930,752	\$409,017,461	\$434,948,214

The 10 busiest round trip routes in terms of number of round trip flights in the quarter. Exclude canceled flights when performing the calculation.:

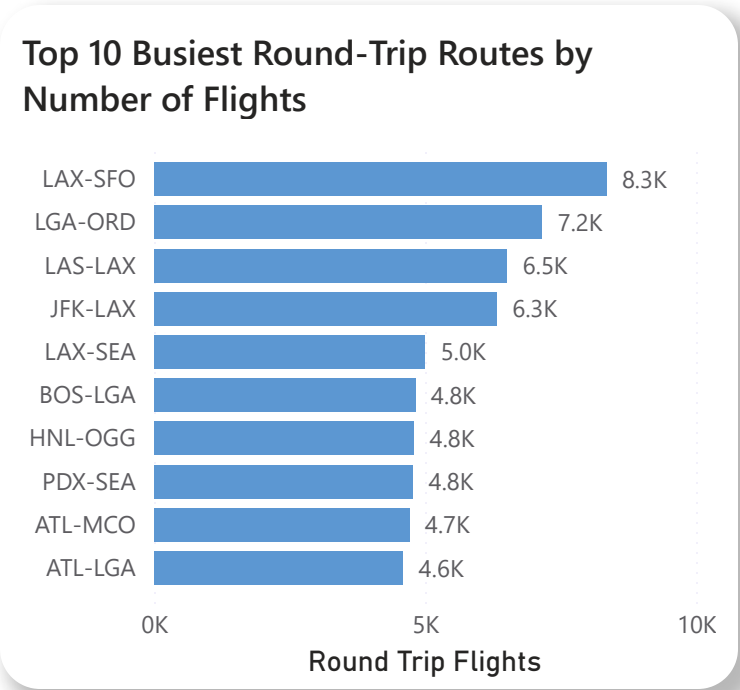
To determine the ten busiest round-trip routes, I began by filtering the dataset to exclude canceled flights, ensuring that only completed operations were counted.

I then normalized each route by pairing flights in both directions between two airports, so that corridors like LAX–SFO and SFO–LAX were treated as a single route. After establishing these route pairs, I aggregated the total number of flights across the quarter and ranked them in descending order.

The top ten routes were then selected and displayed in the chart.

Measures used:

```
Round Trip Flights =  
CALCULATE (  
    COUNTROWS ( 'Flight codes' ),  
    'Flight codes'[CANCELLED] = 0  
)
```



A horizontal bar chart clearly ranks routes by flight volume and keeps route names easy to read.

The horizontal bar chart is the most effective visual for this analysis because it clearly compares flight volumes across multiple routes in a straightforward, side-by-side format.

The length of each bar makes it easy to see which routes dominate, while the labeling of both airports and flight counts provides immediate context.

Unlike a table or pie chart, the bar chart emphasizes relative scale and highlights the contrast between business-heavy corridors and leisure-driven destinations at a glance.

This makes it the best choice for communicating the busiest routes quickly and intuitively.

The 10 most profitable round trip routes (without considering the upfront airplane cost) in the quarter. Along with the profit, show total revenue, total cost, summary values of other key components and total round trip flights in the quarter for the top 10 most profitable routes. Exclude canceled flights from these calculations.

To identify the ten most profitable round-trip routes for the quarter, I first excluded all canceled flights to ensure the analysis reflected only completed operations.

I then calculated profitability by subtracting total operating costs from total revenue, without considering the upfront airplane purchase cost. Each route was normalized to account for flights in both directions, and I aggregated key financial metrics: total revenue, total cost, and total profit.

In addition, I summarized other relevant components and included the total number of round-trip flights for each route. Finally, I ranked the routes by profit and selected the top ten.

The grouped bar chart is the most effective visual for this analysis because it allows direct comparison of revenue, cost, and profit across multiple routes in a single view.

By using distinct colors for each financial metric, the chart highlights the balance between revenue generation and cost efficiency, making it easy to see which routes deliver the highest margins. Unlike a table or single-metric chart, this format emphasizes both scale and profitability drivers, providing a comprehensive financial snapshot at a glance.

A clustered bar chart to compare revenue, cost, and profit side-by-side for the top 10 routes, making profitability drivers and differences across routes easy to interpret.

Measures used:

```
Baggage Revenue =  
CALCULATE (  
    SUMX (Flight codes',  
        200  
        * 'Flight codes'[OCCUPANCY_RATE]  
        * 0.5  
        * 35), 'Flight codes'[CANCELLED] = 0)
```

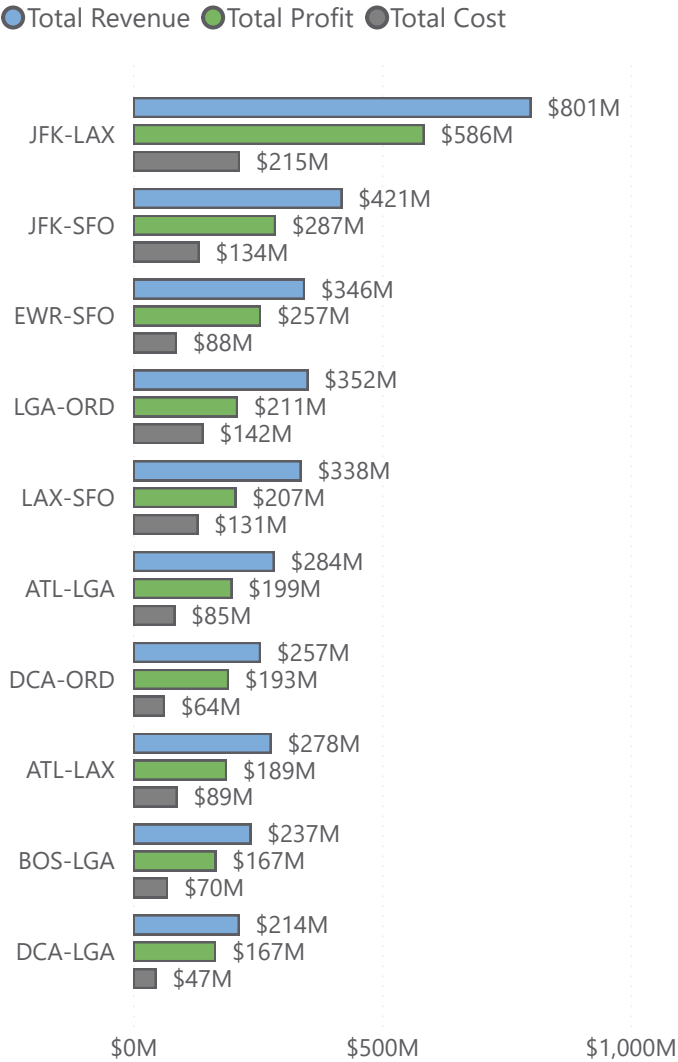
```
Ticket Revenue =  
    [Total Passengers] * [Average Ticket Fare]
```

```
Total Revenue =  
    [Ticket Revenue] + [Baggage Revenue]
```

```
Total Cost =  
    [Distance Cost] + [Airport Cost]+ [Departure Delay Cost]+  
    [Arrival Delay Cost]
```

```
Round Trip Flights =  
CALCULATE (  
    COUNTROWS ( 'Flight codes' ),  
    'Flight codes'[CANCELLED] = 0)
```

Top 10 Most Profitable Round-Trip Routes (Revenue vs Cost vs Profit)



The 5 round trip routes that you recommend to invest in based on any factors that you choose.

When evaluating potential investment routes, focusing solely on total profit can be misleading because it may favor routes with high overall revenue but low efficiency.

By calculating profit per flight, we measure how much value each individual operation contributes, which highlights efficiency and operational sustainability.

Pairing this with flight volume ensures that we are not only identifying efficient routes but also those with sufficient scale to make the investment worthwhile.

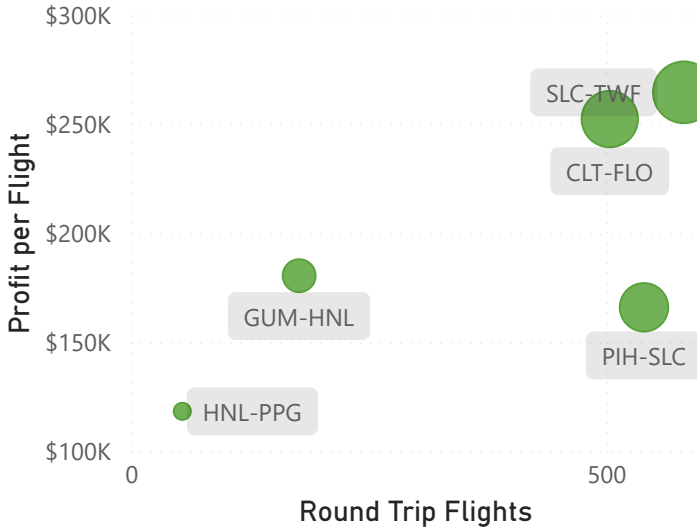
Measures used:

Breakeven Round Trip Flights =
DIVIDE ([Upfront Airplane Cost], [Profit per Flight])

Profit per Flight =
DIVIDE ([Total Profit], [Round Trip Flights])

Round Trip Flights =
CALCULATE (COUNTRROWS ('Flight codes'), 'Flight codes'[CANCELLED] = 0)

Top 5 Recommended Round-Trip Routes: Profitability vs Flight Volume



An Investment routes using a scatter analysis that balances profitability, efficiency, and scale.

This dual metric balances two critical dimensions:

- **Efficiency (profit per flight):** Ensures that each flight is financially viable and not dependent on sheer volume to mask inefficiencies.
- **Scale (flight volume):** Confirms that the route has enough demand to sustain profitability over time and justify resource allocation.

Together, these metrics provide a more holistic view than revenue or profit alone.

A route with high profit per flight but very few flights may lack scalability, while a route with high volume but low profit per flight may drain resources.

By combining both, we identify routes that are both financially efficient and operationally scalable, making them the strongest candidates for investment.

The number of round trip flights it will take to breakeven on the upfront airplane cost for each of the 5 round trip routes that you recommend. Print key summary components for these routes.

Aircraft Investment Breakeven by Recommended Round-Trip Route

Route	Profit per Flight	Breakeven Round Trip Flights	Total Cost	Total Profit	Total Revenue
SLC-TWF	\$264,533	378	\$6,284,033	\$153,958,122	\$160,242,155
CLT-FLO	\$252,272	396	\$4,681,395	\$127,145,276	\$131,826,671
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PIH-SLC	\$165,918	603	\$5,260,380	\$89,595,833	\$94,856,213
HNL-PPG	\$118,204	846	\$1,714,076	\$6,383,028	\$8,097,105
Total	\$981,352	2777	\$25,930,752	\$409,017,461	\$434,948,214

I used a table here because the breakeven analysis requires precise, comparable values across several financial metrics. With only five recommended routes, a table provides the clearest and most accurate way to evaluate capital recovery.

To determine the number of round-trip flights required to breakeven on the upfront airplane cost for each of the five recommended routes, I first calculated the profit per flight by subtracting total operating costs from total revenue. Using this figure, I divided the upfront airplane investment cost by the profit per flight to identify the breakeven point in terms of flights. This calculation shows how many flights must be operated before the initial aircraft investment is fully recovered.

The table summarizes key components for each route, including profit per flight, breakeven round-trip flights, total cost, total profit, and total revenue. For example, the SLC–TWF route requires 378 flights to breakeven, while CLT–FLO requires 396. Routes with lower profit per flight, such as HNL–PPG, require significantly more flights (846) to reach breakeven. This comparison highlights both efficiency and scale, helping decision-makers evaluate which routes justify investment more quickly.

The tabular format is the most effective visual here because it organizes multiple financial metrics side by side, making it easy to compare across routes. Unlike a chart that emphasizes only one dimension, the table provides a comprehensive snapshot of profitability, costs, and breakeven thresholds, which is critical for investment planning.

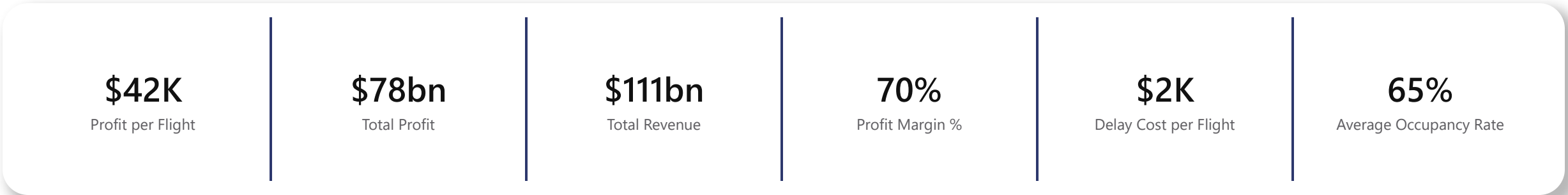
Measures used:

Breakeven Round Trip Flights =
DIVIDE (
 [Upfront Airplane Cost],
 [Profit per Flight])

Round Trip Flights =
CALCULATE (
 COUNTROWS ('Flight codes'),
 'Flight codes'[CANCELLED] = 0)

Profit per Flight =
DIVIDE (
 [Total Profit],
 [Round Trip Flights])

Key Performance Indicators (KPI's) that you recommend tracking in the future to measure the success of the round trip routes that you recommend



The success of the recommended round-trip routes can be measured through a balanced set of KPIs that capture profitability, efficiency, and demand.

Profit per flight highlights the efficiency of each operation, while **total profit** and **total revenue** show the overall financial impact and market strength of the routes.

The **profit margin percentage** provides a clear measure of how effectively revenue is converted into profit, ensuring strong cost control.

Operational reliability is tracked through **delay cost per flight**, which quantifies the financial impact of disruptions, and the **average occupancy rate**, which reflects how well capacity is being utilized. Together, these six KPIs provide a comprehensive view of route performance, ensuring that investment decisions are guided not only by financial returns but also by operational efficiency and demand sustainability.

- **Profit per Flight (\$42K)** – Measures efficiency at the flight level, ensuring each trip contributes positively to profitability.
- **Total Profit (\$78bn)** – Captures the overall financial impact of all routes, showing their contribution to company earnings.
- **Total Revenue (\$111bn)** – Reflects demand strength and pricing effectiveness, highlighting market growth opportunities.
- **Profit Margin % (70%)** – Indicates how efficiently revenue is converted into profit, allowing benchmarking against industry standards.
- **Delay Cost per Flight (\$2K)** – Quantifies the financial impact of operational delays, highlighting areas where reliability improvements are needed.
- **Average Occupancy Rate (65%)** – Shows how well capacity is being utilized, ensuring aircraft resources are not underused.

Airport_Codes Data:

A ^B _C TYPE			A ^B _C IATA_CODE			A ^B _C NAME		
2 distinct, 0 unique			4453 distinct, 4453 unique			4438 distinct, 4425 unique		
1	medium_airport	WKK	Aleknagik / New Airport					
2	medium_airport	HIR	Honiara International Airport					
3	medium_airport	MUA	Munda Airport					
4	medium_airport	AHJ	Hongyuan Airport					
5	medium_airport	INU	Nauru International Airport					
6	medium_airport	AXF	Alxa Left Banner Bayanhot Airport					
7	medium_airport	BUA	Buka Airport					
8	medium_airport	CMU	Chimbu Airport					
9	medium_airport	DAU	Daru Airport					
10	medium_airport	GKA	Goroka Airport					
11	medium_airport	GUR	Gurney Airport					
12	medium_airport	PNP	Girua Airport					
13	medium_airport	HKN	Kimbe Airport					
14	medium_airport	KMA	Kerema Airport					
15	medium_airport	KVG	Kavieng Airport					
16	medium_airport	MAG	Madang Airport					
17	medium_airport	HGU	Mount Hagen Kagamuga Airport					
18	medium_airport	MDU	Mendi Airport					
19	medium_airport	MAS	Momote Airport					
20	medium_airport	LAE	Nadzab Airport					
21	large_airport	POM	Port Moresby Jacksons International Airport					
22	medium_airport	RAB	Tokua Airport					
23	medium_airport	VAI	Vanimo Airport					
24	medium_airport	WBM	Wapenamanda Airport					
25	medium_airport	WWK	Wewak International Airport					
26	medium_airport	ZXT	Zabrat Airport					
27	medium_airport	BCW	Benguera Island Airport					
28	medium_airport	JAM	Bezmer Air Base					
29	medium_airport	JEG	Aasiaat Airport					
30	medium_airport	UAK	Narsarsuaq Airport					

Description:

- The pipeline begins by reading the raw CSV file Airport_Codes.csv
- This ensures the dataset is consistently imported into Power Query for structured processing.
- Promote Headers The first row is promoted to column headers, giving each field a meaningful name and improving usability in subsequent steps.
- Select Relevant Columns Only the fields TYPE, IATA_CODE, and NAME are retained. This reduces the dataset to the essential attributes needed for airport identification and classification.
- Filter by Airport Type Rows are filtered to include only airports classified as large_airport or medium_airport. This focuses the dataset on significant airports that are most relevant for operational and analytical purposes.
- Filter by Valid IATA Codes Rows where IATA_CODE is null or empty are removed. This ensures that only airports with valid identifiers are included, which is critical for accurate mapping and integration with other datasets.
- Remove Duplicates Duplicate records based on IATA_CODE are eliminated. This guarantees that each airport is represented uniquely, preventing redundancy in reporting and analysis.

Power Query (M Code):

```
let
Source = Csv.Document(File.Contents("D:\PowerBi\Datasets\data\Airport_Codes.csv"),[Delimiter="," , Columns=8,
Encoding=1252, QuoteStyle=QuoteStyle.None]),
#"Promoted Headers" = Table.PromoteHeaders(Source, [PromoteAllScalars=true]),
#"Removed Other Columns" = Table.SelectColumns(#"Promoted Headers",{"TYPE", "IATA_CODE", "NAME"}),
#"Filtered Rows" = Table.SelectRows(#"Removed Other Columns", each ([TYPE] = "large_airport" or [TYPE] =
"medium_airport")),
#"Filtered Rows1" = Table.SelectRows(#"Filtered Rows", each [IATA_CODE] <> null and [IATA_CODE] <> ""),
#"Removed Duplicates" = Table.Distinct(#"Filtered Rows1", {"IATA_CODE"})
in
#"Removed Duplicates"
```

Flights Data:

	FL_DATE	ORIGIN_AIRPORT_ID	ORIGIN	DEST_AIRPORT_ID	DESTINATION	1.2 DEP_DELAY	1.2 ARR_DELAY	1.2 CANCELLED	1.2 AIR_TIME	1.2 DISTANCE	1.2 OCCUPANCY_RATE
1	3/2/2019	14635	RSW	11042	CLE	-8	-6	0	143	1025	0.97
2	3/2/2019	14635	RSW	11066	CMH	1	5	0	135	930	0.55
3	3/2/2019	14635	RSW	11066	CMH	0	4	0	132	930	0.91
4	3/2/2019	14635	RSW	11066	CMH	11	14	0	136	930	0.67
5	3/2/2019	14635	RSW	11259	DAL	0	-17	0	151	1005	0.62
6	3/2/2019	14635	RSW	11986	GRR	-2	-8	0	162	1147	0.49
7	3/2/2019	14635	RSW	12339	IND	11	17	0	143	945	0.49
8	3/2/2019	14635	RSW	12339	IND	-5	8	0	139	945	0.37
9	3/2/2019	14635	RSW	12339	IND	-9	-16	0	135	945	0.31
10	3/2/2019	14635	RSW	12339	IND	0	11	0	150	945	0.7
11	3/2/2019	14635	RSW	13198	MCI	-7	-5	0	180	1155	0.84
12	3/2/2019	14635	RSW	13232	MDW	5	8	0	164	1105	0.8
13	3/2/2019	14635	RSW	13232	MDW	-2	-4	0	164	1105	0.36
14	3/2/2019	14635	RSW	13232	MDW	-2	11	0	173	1105	0.35
15	3/2/2019	14635	RSW	13232	MDW	1	1	0	166	1105	0.67
16	3/2/2019	14635	RSW	13232	MDW	-1	5	0	164	1105	0.3
17	3/2/2019	14635	RSW	13342	MKE	-4	-6	0	173	1183	0.81
18	3/2/2019	14635	RSW	13342	MKE	-3	-1	0	172	1183	0.94
19	3/2/2019	14635	RSW	14122	PIT	-2	1	0	134	966	0.88
20	3/2/2019	14635	RSW	14122	PIT	-6	-17	0	126	966	0.88
21	3/2/2019	14635	RSW	14307	PVD	19	17	0	152	1200	0.42
22	3/2/2019	14635	RSW	14307	PVD	-5	-13	0	150	1200	0.38
23	3/2/2019	14635	RSW	14576	ROC	-1	11	0	166	1166	0.56
24	3/2/2019	14635	RSW	15016	STL	-3	2	0	150	979	0.7
25	3/2/2019	14635	RSW	15016	STL	10	7	0	137	979	0.99
26	3/2/2019	14635	RSW	15016	STL	8	14	0	146	979	0.75
27	3/2/2019	14679	SAN	10140	ABQ	0	-19	0	69	628	0.88
28	3/2/2019	14679	SAN	10397	ATL	22	7	0	199	1892	0.58
29	3/2/2019	14679	SAN	10423	AUS	-2	-17	0	127	1164	0.39
30	3/2/2019	14679	SAN	10693	BNA	28	8	0	175	1751	0.8
31	3/2/2019	14679	SAN	10713	BOI	35	22	0	111	749	0.98

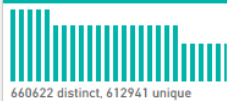
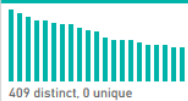
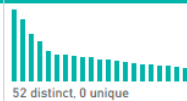
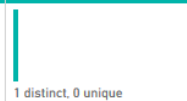
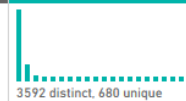
Power Query (M Code):

```
let
Source = Csv.Document(File.Contents("D:\PowerBi\Datasets\data\Flights.csv"),[Delimiter=",", Columns=16, Encoding=1252, QuoteStyle=QuoteStyle.None]),
#"Promoted Headers" = Table.PromoteHeaders(Source, [PromoteAllScalars=true]),
#"Removed Columns" = Table.RemoveColumns(#"Promoted Headers",{"OP_CARRIER", "TAIL_NUM", "OP_CARRIER_FL_NUM"}),
#"Split Column by Delimiter" = Table.SplitColumn(#"Removed Columns", "ORIGIN_CITY_NAME", Splitter.SplitTextByDelimiter(",", QuoteStyle.Csv), {"ORIGIN_CITY", "ORIGIN_STATE"}),
#"Split Column by Delimiter1" = Table.SplitColumn(#"Split Column by Delimiter", "DEST_CITY_NAME", Splitter.SplitTextByDelimiter(",", QuoteStyle.Csv), {"DEST_CITY", "DEST_STATE"}),
#"Filtered Rows" = Table.SelectRows(#"Split Column by Delimiter1", each ([CANCELLED] = "0.0")),
#"Changed Type" = Table.TransformColumnTypes(#"Filtered Rows",{{"FL_DATE", type date}, {"DEP_DELAY", type number}, {"ARR_DELAY", type number}, {"AIR_TIME", type number}, {"DISTANCE", type number}, {"OCCUPANCY_RATE", type number}, {"CANCELLED", type number}}),
#"Removed Columns1" = Table.RemoveColumns(#"Changed Type",{"ORIGIN_CITY", "ORIGIN_STATE", "DEST_CITY", "DEST_STATE"}),
#"Removed Errors" = Table.RemoveRowsWithErrors(#"Removed Columns1", {"AIR_TIME"}),
#"Removed Errors1" = Table.RemoveRowsWithErrors(#"Removed Errors", {"DISTANCE"})
in
#"Removd Errors1"
```

Description:

- The process begins by loading the raw CSV file Flights.csv
- This ensures the dataset is consistently imported into Power Query for structured processing.
- Promote Headers The first row is promoted to column headers, giving each field a meaningful name and improving usability in subsequent steps.
- Remove Irrelevant Columns The fields OP_CARRIER, TAIL_NUM, and OP_CARRIER_FL_NUM are removed. These identifiers are not required for the analysis, which streamlines the dataset and reduces unnecessary detail.
- Split City/State Columns The ORIGIN_CITY_NAME column is split into ORIGIN_CITY and ORIGIN_STATE, and the DEST_CITY_NAME column is split into DEST_CITY and DEST_STATE. This provides cleaner separation of city and state information, making it easier to analyze geographic patterns.
- Filter Rows Only records where CANCELLED = "0.0" are retained. This ensures the dataset focuses on completed flights, excluding cancelled trips that could distort delay and occupancy metrics.
- Change Data Types Several columns are converted to appropriate data types:
 - FL_DATE → Date
 - DEP_DELAY, ARR_DELAY, AIR_TIME, DISTANCE, OCCUPANCY_RATE, CANCELLED → NumberThis enforces consistency and ensures accurate calculations for time, distance, and occupancy metrics.
- Remove Temporary Split Columns After splitting, the intermediate columns (ORIGIN_CITY, ORIGIN_STATE, DEST_CITY, DEST_STATE) are removed. This keeps the dataset concise while retaining the cleaned structure.
- Remove Rows with Errors Any rows containing errors in AIR_TIME or DISTANCE are eliminated. This step safeguards data quality by ensuring invalid records do not distort performance metrics.

Tickets Data:

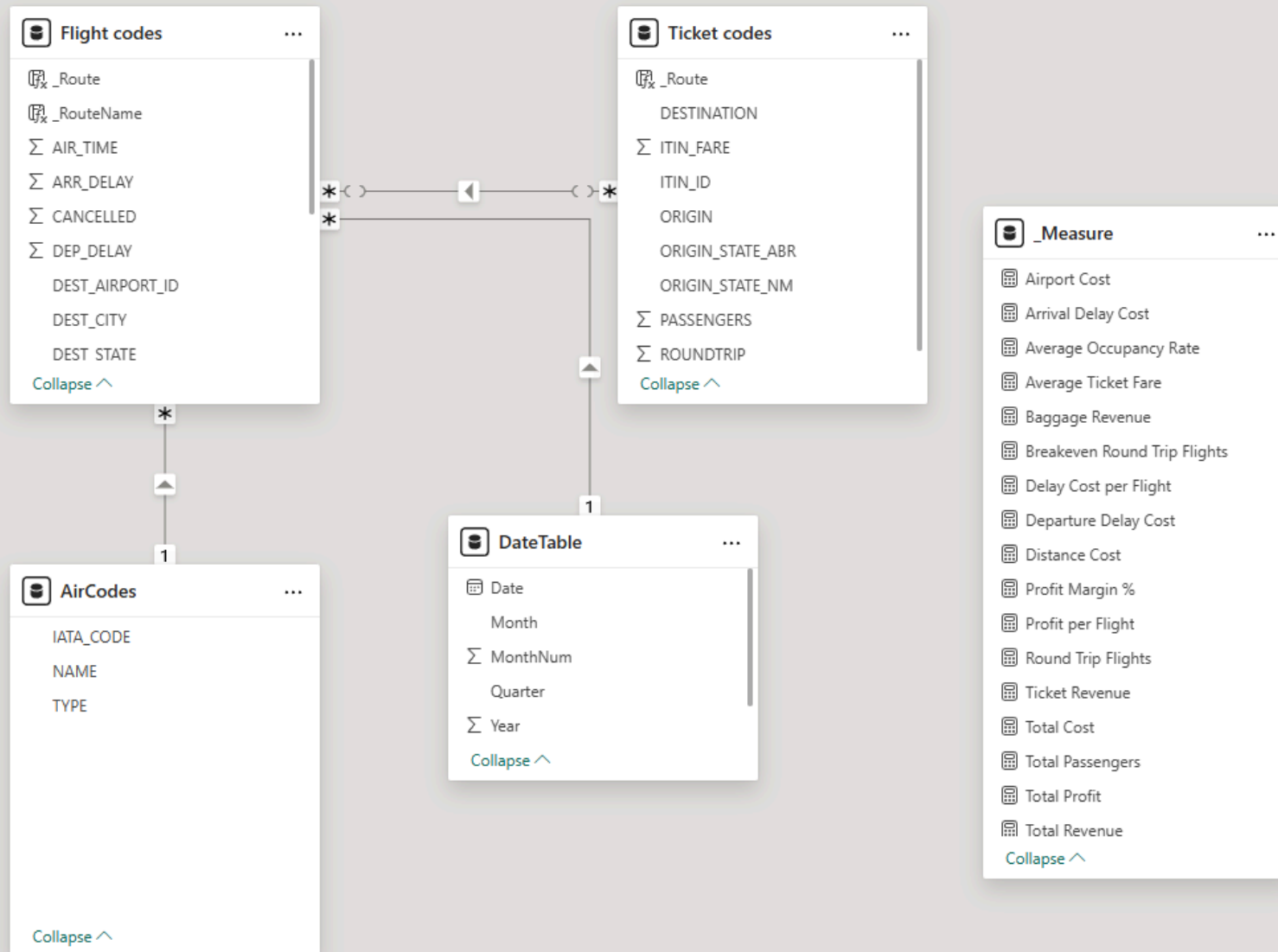
	A ^B _C ITIN_ID	A ^B _C ORIGIN	A ^B _C ORIGIN_STATE_ABR	A ^B _C ORIGIN_STATE_NM	1.2 ROUNDTRIP	1 ² ₃ PASSENGERS	\$ ITIN_FARE	A ^B _C DESTINATION
								
1	201912723049	ABI	TX	Texas		1	736.00	DAB
2	201912723085	ABI	TX	Texas		1	570.00	COS
3	201912723491	ABI	TX	Texas		1	564.00	MCO
4	201912723428	ABI	TX	Texas		1	345.00	LGA
5	201912723337	ABI	TX	Texas		1	1,647.00	JAX
6	201912723447	ABI	TX	Texas		1	425.00	UT
7	201912723815	ABI	TX	Texas		1	1,445.00	FAI
8	201912722986	ABI	TX	Texas		1	842.00	BNA
9	201912723745	ABI	TX	Texas		1	279.00	RDU
10	201912723032	ABI	TX	Texas		1	379.00	CLE
11	201912723486	ABI	TX	Texas		1	934.00	MCO
12	201912723723	ABI	TX	Texas		1	644.00	PIT
13	201912723128	ABI	TX	Texas		1	469.00	PVD
14	201912723519	ABI	TX	Texas		1	889.00	MIA
15	201912723259	ABI	TX	Texas		1	645.00	HOU
16	201912722918	ABI	TX	Texas		1	607.00	ASE
17	201912723704	ABI	TX	Texas		1	675.00	PHX
18	201912723031	ABI	TX	Texas		1	11.00	CLE
19	201912723396	ABI	TX	Texas		1	323.00	LAX
20	201912723793	ABI	TX	Texas		1	528.00	SAV
21	201912723516	ABI	TX	Texas		1	507.00	MIA
22	201912723394	ABI	TX	Texas		1	239.00	LAX
23	201912723072	ABI	TX	Texas		1	867.00	DCA
24	201912723174	ABI	TX	Texas		1	544.00	ECP
25	201912722925	ABI	TX	Texas		1	872.00	CLT
26	201912723407	ABI	TX	Texas		1	874.00	LAX
27	201912723768	ABI	TX	Texas		1	277.00	SAN
28	201912723029	ABI	TX	Texas		1	489.00	CID
29	201912723472	ABI	TX	Texas		1	474.00	MCO
30	201912723672	ABI	TX	Texas		1	592.00	BDL

Description:

- The pipeline begins by reading the raw CSV file Tickets.csv
- This ensures the dataset is consistently imported into Power Query for further processing.
- Promote Headers The first row is promoted to column headers, giving each field a meaningful name and improving usability in subsequent steps.
- Remove Irrelevant Columns Columns YEAR, QUARTER, ORIGIN_COUNTRY, and REPORTING_CARRIER are removed. These fields are not required for the current analysis, which streamlines the dataset and reduces noise.
- Filter Rows Only records where ROUNDTRIP = "1.0" are retained. This focuses the dataset on roundtrip tickets, aligning with the business requirement to analyze roundtrip travel behavior and fare structures.
- Change Data Types
 - ITIN_FARE is converted to Currency for accurate financial calculations.
 - PASSENGERS is converted to Int64 to ensure whole-number passenger counts.
 - ROUNDTRIP is converted to Number for proper numeric comparisons. These conversions enforce consistency and prevent calculation errors.
- Remove Rows with Errors Any rows containing errors in ITIN_FARE are eliminated. This step safeguards data quality by ensuring invalid or corrupted records do not distort reporting or analysis.

Power Query (M Code):

```
let
Source = Csv.Document(File.Contents("D:\PowerBi\Datasets\data\Tickets.csv"),[Delimiter="," , Columns=12, Encoding=1252, QuoteStyle=QuoteStyle.None]),
#"Promoted Headers" = Table.PromoteHeaders(Source, [PromoteAllScalars=true]),
#"Removed Columns" = Table.RemoveColumns(#"Promoted Headers",{YEAR, "QUARTER", "ORIGIN_COUNTRY", "REPORTING_CARRIER"}),
#"Filtered Rows" = Table.SelectRows(#"Removed Columns", each ([ROUNDTRIP] = "1.0")),
#"Changed Type" = Table.TransformColumnTypes(#"Filtered Rows",{{"ITIN_FARE", Currency.Type}, {"PASSENGERS", Int64.Type}, {"ROUNDTRIP", type number}}),
#"Removed Errors" = Table.RemoveRowsWithErrors(#"Changed Type", {"ITIN_FARE"})
in
#"Removed Errors"
```



- **Flight Codes** Contains operational flight data such as route identifiers, delays (arrival and departure), cancellations, air time, destination airport details, and city/state information. *Purpose:* Used to analyze flight performance and operational efficiency.
- **Ticket Codes** Holds ticket-level information including origin, destination, fare, passenger counts, roundtrip indicators, and itinerary IDs. *Purpose:* Supports revenue, passenger volume, and fare analysis.
- **AirCodes** Provides airport reference data with IATA codes, airport names, and classifications (e.g., large or medium airports). *Purpose:* Acts as a lookup table for mapping flight and ticket data to specific airports.
- **DateTable** A standard calendar table with fields for date, month, quarter, and year. *Purpose:* Enables time-based analysis, trend reporting, and period comparisons.
- **_Measure** A calculation table containing KPIs and metrics such as ticket revenue, total passengers, profit margin, delay costs, and breakeven analysis. *Purpose:* Centralizes business measures for reporting and dashboards.

Relationships

- Flight and ticket tables are linked through route identifiers, allowing integration of operational and financial perspectives.
- AirCodes connects via IATA codes to provide airport context.
- DateTable links to flight and ticket dates for time-series analysis.
- The _Measure table aggregates across these relationships to produce business insights.

Analytical Use

This model enables:

- **Operational Analysis:** Flight delays, cancellations, air time, and distance efficiency.
- **Financial Analysis:** Ticket revenue, fare averages, occupancy rates, and profitability.
- **Geographic Analysis:** Mapping flights to origin/destination airports and states.
- **Time-Based Analysis:** Trends across months, quarters, and years.
- **Strategic KPIs:** Profit margin %, breakeven flights, and cost breakdowns.